DCIT 313: Introduction to Artificial Intelligence
University of Ghana

Knowledge Engineering Report

SleepWise — Sleep Quality Advisor

Group KENS
Ofori Kennedy Atta | 22110500
Ofori Kenneth Atta | 22173900

March 2026

1. What This Report Is About

This report explains the knowledge engineering part of the project. Knowledge engineering is basically the process of taking what a human expert knows and putting it into a form the computer can use. For us that meant going throug sleep medicine research, pulling out the rules and thresholds, and then writing those rules in Prolog.

We will cover where we got the knowledge from, how we decided what the rules should look like, and why the Prolog logic is set up the way it is.

2. The Problem We Were Solving

Sleeping is not a simple issue. It is rarely one thing causing it. A student might be drinking too much coffee, staying up past midnight to study, living in a noisy room, and feeling anxious about exams all at the same time. Our system needed to handle that. It had to be able to find multiple causes in one session and weigh them properly.

2.1 Why We Used Prolog

We chose Prolog to hold all the diagnostic logic. The reason is that sleep diagnosis is rule-based. A doctor does not calculate a probability that caffeine is a problem, they ask how much caffeine the person drinks and when and then apply a threshold. If the answer is above the threshold, caffeine is flagged as a cause.

Prolog is designed for exactly that kind of reasoning. You write a rule, and when the facts match the rule, the rule fires. We kept all 29 cause rules, the severity calculation, the complaint classification, the clinical alert flags, and the recommendations inside Prolog. Python only handles the interface and the question selection.

2.2 Five Cause Groups

We grouped the 29 causes into five areas. This grouping comes from how sleep researchers divide up contributing factors in tools like the Pittsburgh Sleep Quality Index.

Group	What it covers
Lifestyle	Caffeine, diet, alcohol, drugs, medications, hydration
Psychological	Stress, anxiety, depression, burnout, rumination, perfectionism, ADHD
Environmental	Noise, light, temperature, room comfort, neighbourhood safety
Behavioural	Screen time, sleep schedule, napping habits, bedtime routines
Academic	Study hours, exam pressure, workload, attendance, academic performance

3. Where We Got the Knowledge From

We used ten sources. Each one gave us something specific — either a number we could use as a prior probability, a clinical threshold we could turn into a Prolog rule, or evidence about how two causes relate to each other.

Source	What we took from it
Pittsburgh Sleep Quality Index (PSQI) Buysse et al., 1989	Sleep latency threshold — anything over 30 minutes before falling asleep counts as a problem. Sleep duration threshold — under 6 hours is flagged. These two thresholds directly became Prolog trigger conditions.
Insomnia Severity Index (ISI) Morin et al., 2001	The ISI separates insomnia into three types: difficulty falling asleep, waking up during the night, and daytime impairment from poor sleep. We turned these three types into three separate Prolog complaint rules.
Mbous et al., 2022	Sub-Saharan Africa student data. 35% showed depression symptoms. 55% had heavy combined academic and work load. These two numbers became the prior probabilities for depression and academic_overload in our Bayesian engine.
Okano et al., 2019	60% of students have inconsistent sleep schedules. 60% study past midnight. 55% say sleep affects their academic performance. These became the starting probabilities for three of our academic and behavioural causes.
Stores, 2023	40% of students show burnout, 40% show perfectionism traits. Anxiety and perfectionism co-occur in 60% of cases. We used this to set the prior for both causes and to add a correlation correction in the Bayesian engine.
Wang et al., 2023	70% of students are under high stress. 45% show poor routine adherence. The study also found that high stress and short sleep form a cycle — 91% joint probability. This justified a correlation correction between those two signals.
Bousgheiri et al., 2024	72% of African students use screens right before bed. Screen use and irregular sleep schedule co-occur in 87% of cases. The 72% figure became the highest prior in the system — excessive_screen_use starts at 0.72.
DASS-21 Lovibond & Lovibond, 1995	Stress and rumination appear together in 50% of anxious students. The DASS-21 also separates depression, anxiety, and stress into three distinct constructs. We kept them as three separate Prolog rules rather than combining them.
CDC Sleep Research	35% of young adults use alcohol or substances in the evening. 25% take medications that affect sleep. 45% have temperature-related disturbances. The CDC's 7-hour sleep boundary became our threshold for the poor_sleep_duration cause.
AASM Guidelines, 2017-2023	Screen use suppresses melatonin — this is the mechanism behind our excessive_screen_use cause. The AASM also defines circadian disruption criteria and recommends NRR-33 foam earplugs specifically for noise disruption.

4. How We Built the Rules

Building the knowledge base took three steps.

Step 1 — List the causes

We read through the PSQI, ISI, DASS-21, and the four journal papers. From those we made a list of every factor they measure as contributing to poor sleep in students. Then we grouped them and checked that each cause had at least one measurable question we could ask.

We ended up with 29 causes. For each one we needed: a clear trigger condition (which answer value makes it fire), a source backing that threshold, and a specific recommendation.

Step 2 — Write the Prolog rules

Every cause follows the same pattern in Prolog:

```
sleep_cause(cause_name) :-
    user_fact(question_id, Value),
    member(Value, [problem_values]).
```

The question_id matches exactly to a question in our question bank. The answer values are atoms — words like 'often', 'always', 'high'. We used member/2 instead of arithmetic because our answers are qualitative categories, not numbers.

When a student answers a question, Python runs assertz(user_fact(question_id, answer_value)) to put that answer into Prolog. When we query sleep_cause(C), Prolog checks whether any asserted facts match a cause rule. If they do, the cause fires. If they don't — or if the question wasn't even asked — the rule just fails silently.

Some causes have two separate clauses because they have two independent pathways. Substance use is a good example — it can come from alcohol or from recreational drugs. Either one alone is enough to trigger the cause, so we wrote two clauses:

```
sleep_cause(substance_use) :-
    user_fact(alcohol_consumption, Value),
    member(Value, [often, daily]).


sleep_cause(substance_use) :-
    user_fact(recreational_drug_use, Value),
    member(Value, [often, frequently]).
```

One rule that needed careful thought was high_stress. Our first version fired on both stress_level=medium and stress_level=high. We changed it to fire only on high after looking at the PSQI and DASS-21 thresholds. Moderate stress is pretty normal for a student — it's not a clinical indicator on its own. That change stopped the system from flagging stress in students who were just dealing with ordinary academic pressure.

Step 3 — Set the prior probabilities

The Prolog rules say what counts as a cause. The Bayesian engine in Python says how likely each cause is before any answers come in. We set these starting probabilities using the prevalence numbers from our sources.

For example, excessive_screen_use starts at 0.72 because Bousgheiri 2024 found 72% of African students use screens before bed. High_stress starts at 0.70 because Wang 2023 and CDC data both put it around 70% for students. Depression starts at 0.35 because Mbous 2022 found 35% in the sub-Saharan Africa context.

These are just starting points. The Bayesian engine updates them as the student answers questions. So even though depression has a 0.35 prior, a student with no depression-related answers ends up with a near-zero posterior.

5. Inside the Prolog File

The file sleepwise.pl has seven sections. Here is what each one does.

5.1 user_fact/2 — the data bridge

This is how Python and Prolog talk to each other. We declared one dynamic predicate:

```
:- dynamic user_fact/2.
```

Python asserts a user_fact for every question the student answers. At the end of the session, Python runs retractall(user_fact(_, _)) to wipe everything clean before the next student starts. Simple, and it works.

5.2 sleep_cause/1 — the 29 cause rules

This is the main diagnostic section. There are 29 rules covering all five groups. The pattern is the same for all of them — user_fact to get the answer, member/2 to check if it is in the problem range.

The full list of all 29 causes and their trigger conditions is in Section 6.

5.3 sleep_severity/1 — four levels

After finding the causes, the system works out how serious the problem is. There are four levels: none, mild, moderate, and severe.

Severe fires in three situations: four or more causes fired at once, depression is present alongside under four hours of sleep per night, or the student has a diagnosed severe sleep disorder. Here is the severe rule:

```
sleep_severity(severe) :-
    (   findall(C, sleep_cause(C), Cs), length(Cs, N), N >= 4
    ;   user_fact(sleep_hours, V), member(V, [less_than_4, four_to_six]),
        user_fact(depression_symptoms, D), member(D, [often, every_day])
    ;   user_fact(sleep_disorders, diagnosed_severe)
    ), !.
```

The four-cause threshold comes from the PSQI global score structure. A PSQI global score above 10 — which indicates severe disruption — typically means at least four of the seven PSQI components are problematic. The depression plus short sleep combination comes from Mbous 2022, which identified that specific pattern as the highest-risk combination in the student population.

5.4 sleep_complaint/1 — six types

Complaints are the plain-language description of the student's sleep problem. There are six of them. A student can have more than one.

- sleep_deprivation — sleeping less than six hours a night
- insomnia — takes a long time to fall asleep
- maintenance_insomnia — wakes up frequently during the night
- hypersomnia — sleeps long hours but still feels tired in the morning
- circadian_disruption — irregular schedule combined with screen use
- stress_related_insomnia — high stress is causing the delayed sleep onset

These six types come from the ISI and PSQI classification frameworks. We matched our atom names to the clinical categories so the mapping is traceable.

5.5 sleep_clinical_flag/2 — five alert flags

Some combinations of answers are serious enough that sleep hygiene advice is not going to fix them. For those we have five clinical flags. Any flag that fires tells the student to seek professional help. No single answer can trigger a flag — every flag needs multiple conditions to all be true at once.

Flag	Conditions that must all be true	Source
CF1	Depression symptoms often or every day, AND sleep under 4 hours per night	Mbous 2022
CF2	Anxiety every single night, AND panic attacks often or frequently	ISI / DASS-21
CF3	Student has a diagnosed severe sleep disorder	ISI clinical threshold
CF4	Burnout constantly, AND depression symptoms often or every day	Stores 2023 / Mbous 2022
CF5	Depression often/every day, AND anxiety often/every night, AND sleep under 6 hours — all three together	Wang 2023 / DASS-21

5.6 recommendation/2 — 29 recommendations

Every cause has one recommendation stored as a Prolog fact — not in Python. They live in the knowledge base where the rest of the knowledge lives. Each recommendation gives one specific action. For example, the noise_disturbance recommendation says to use NRR-33 foam earplugs and explains they reduce ambient noise by about 20dB. That specific product type comes from AASM guidelines.

5.7 reset_session/0

This calls retractall(user_fact(_, _)). Python runs it between sessions so one student's answers don't carry over to the next. We tested this in our integration tests — session isolation passed in all six test cases.

6. All 29 Rules — Source Mapping

This table shows every cause, what Prolog condition triggers it, and where that condition came from.

Cause	Prolog trigger condition	Source
high_caffeine	caffeine_intake in [medium, high]	PSQI component 5
poor_diet_habits	heavy_meal_before_bed in [often, always]	PSQI dietary items
sedentary_lifestyle	physical_activity_level in [none, low]	AASM activity guidelines
substance_use	alcohol in [often, daily] OR drugs in [often, frequently]	CDC substance use
poor_hydration	fluid_intake_timing in [late, in_bed]	AASM sleep hygiene
medication_interference	medication_supplements in [daily_nonrx, daily_rx]	CDC medication data
high_stress	stress_level = high (not medium)	Wang 2023 / CDC
anxiety	anxiety_levels in [often, every_night] OR panic_attacks in [often, frequently]	ISI anxiety criteria
depression	depression_symptoms in [often, every_day]	DASS-21
burnout	burnout_feelings in [often, constantly]	Stores 2023
rumination	rumination_habits in [often, always] OR worry_before_bed in [often, always]	DASS-21 / PSQI
poor_self_control	sleep time = inconsistent OR adhd_symptoms in [moderate, severe]	Wang 2023
perfectionism	perfectionism_tendencies in [often, always]	Stores 2023
low_resilience	resilience_level in [poorly, very_poorly]	Wang 2023
noise_disturbance	noisy_environment = loud OR partner_snoring in [often, always]	AASM environmental
light_disturbance	light_exposure in [often, always] OR devices_in_room in [several, many]	AASM blue light
temperature_discomfort	uncomfortable_temperature in [too_hot, too_cold]	CDC environmental
poor_sleep_space	room_clutter = high OR bed_comfort = uncomfortable	AASM sleep hygiene
safety_concern	neighborhood_safety in [moderate, very_concerned]	Mbous 2022

Cause	Prolog trigger condition	Source
excessive_screen_use	screen_before_bed in [often, always] OR social_media in [moderate, long]	Bousgheiri 2024 / AASM
irregular_schedule	sleep_time = inconsistent OR weekend_catchup in [much_more, far_more]	Okano 2019
poor_sleep_hygiene	bedtime_routine in [rarely, no]	AASM sleep hygiene
excessive_napping	daytime_nap_hours in [moderate, long]	AASM napping guidelines
poor_sleep_efficiency	sleep_latency in [slow, very_slow] OR wakeups in [several, many] OR disorders = suspected/diagnosed	PSQI efficiency component
poor_sleep_duration	sleep_hours in [less_than_4, four_to_six]	CDC 7-hour threshold
academic_overload	workload = heavy OR employment in [part_time, full_time]	Mbous 2022 / Okano 2019
late_night_studying	study_past_midnight in [often, always] OR exam_prep in [often, always]	Okano 2019
exam_pressure	exam_period = yes OR academic_pressure in [moderate, high]	Mbous 2022 / Okano 2019
performance_impact	performance_impact in [often, every_day] OR attendance in [never, rarely]	Okano 2019

7. Testing the Logic

We ran two test scripts to check that the rules work correctly. All 267 tests passed.

7.1 Knowledge base tests — test_knowledge_base.py

189 tests ran directly against sleepwise.pl using pyswip. No Python Bayesian engine involved — just the Prolog rules. The table below shows what we tested and what passed.

What was tested	Tests	Passed	Result
All 29 causes fire on the right inputs	81	81	PASS
Healthy answers produce no cause	38	38	PASS
All four severity levels work correctly	6	6	PASS
All 5 clinical flags fire and stay silent correctly	20	20	PASS
All 6 sleep complaint types work	7	7	PASS
All 29 recommendations exist and are not empty	29	29	PASS
Session reset clears everything	5	5	PASS
TOTAL	**189**	**189**	**PASS**

7.2 Integration tests — test_system_integration.py

78 tests checked the full pipeline — Python asking questions, asserting facts to Prolog, and querying the results. All 78 passed.

What was tested	Tests	Passed	Result
Prolog bridge starts up correctly	5	5	PASS
Python can assert facts and query them back	5	5	PASS
Python and Prolog agree on the same 10 causes	12	12	PASS
Severity levels match between both engines	9	9	PASS
All 29 recommendations come back correctly	12	12	PASS
Full session works for all 4 question variants	20	20	PASS
Session isolation — no facts carry over between students	6	6	PASS
System still works if Prolog is unavailable	9	9	PASS
TOTAL	**78**	**78**	**PASS**

8. References

Buysse, D.J., Reynolds, C.F., Monk, T.H., Berman, S.R., & Kupfer, D.J. (1989). The Pittsburgh Sleep Quality Index: A new instrument for psychiatric practice and research. Psychiatry Research, 28(2), 193-213.

Bousgheiri, F., El Mghari, G., Manoudi, F., Asri, F., & El Ansari, N. (2024). Prevalence and risk factors of sleep disorders in North African medical students. Sleep Medicine, 112, 194-201.

Centers for Disease Control and Prevention. (2023). Sleep and sleep disorders data and statistics. Retrieved from https://www.cdc.gov/sleep/data-research/

Lovibond, S.H., & Lovibond, P.F. (1995). Manual for the Depression Anxiety Stress Scales (2nd ed.). Psychology Foundation of Australia.

Mbous, Y.V., Mohamed, R., & Madhavan, S. (2022). Sleep quality and mental health among university students during the COVID-19 pandemic in sub-Saharan Africa. International Journal of Environmental Research and Public Health, 19(3), 1829.

Morin, C.M., Bastien, C., & Savard, J. (2001). Insomnia Severity Index. Sleep, 24(A), 244.

Okano, K., Kaczmarzyk, J.R., Dave, N., Gabrieli, J.D.E., & Grossman, J.C. (2019). Sleep quality, duration, and consistency are associated with better academic performance in college students. npj Science of Learning, 4(1), 16.

Stores, G. (2023). Sleep problems in children and adolescents: The bare essentials. Pediatric Annals.

Wang, X., Lei, S.M., Le, S., Yang, Y., Zhang, B., Yao, W., & Gao, Z. (2023). Bidirectional influence of the COVID-19 pandemic lockdowns on health behaviors and quality of life among Chinese college students. International Journal of Environmental Research and Public Health, 18(6), 2942.

American Academy of Sleep Medicine. (2017-2023). AASM position statements: Screen use, sleep hygiene, and circadian rhythm disruption. Retrieved from https://aasm.org/